

Q17	If $\frac{d(f(x))}{dx} = ax + b$, and $f(0) = 0$, then $f(x)$ is equal to a) $a+b$ b) $\frac{ax^2}{2} + bx$ c) $\frac{x^2}{2} + bx + c$ d) b	
Q18.	If $y^2(2-x) = x^3$ then $\frac{dy}{dx}$ at $(1,1)$ is (a) -2 (b) 2 (c) 3 (d) $\frac{-3}{2}$	
	Assertion-Reason based questions In the following questions, a statement of assertion (A) is followed by a statement of Reason (R). Choose the correct answer out of the following choices. (a) both A and R are true and R is the correct explanation of A. (b) both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true.	
Q19	Assertion (A): The domain of $\operatorname{cosec}^{-1} x$ is $(-\infty, -1] \cup [1, \infty)$ Reason (R): $\operatorname{cosec}^{-1}(-2) = \frac{3\pi}{4}$	1
Q20.	(A) The derivative of x^x with respect to x is $x^x(1+\log x)$ (R): The derivative of x^x with respect to x is $(x \cdot x^{x-1})$	1
	Section-B	
Q21.	Show that the function $f: N \rightarrow N$, given by $f(x) = 2x$ is one-one but not onto.	2
Q22.	If $f(x) = \begin{cases} x^2, & x \geq 1 \\ x, & x < 1 \end{cases}$ check whether $f(x)$ is differentiable or not at $x=1$	2
Q23.	Volume of the cube is increasing at the rate of 9c.c./sec. what is the rate at which the surface areas is increasing bwhen the length of an edge of the cube is 12cm. Or Show that the function $f(x) = x^3 - 3x^2 + 6x - 100$ is an increasing function.	2
Q24.	Find $\int \frac{\sin 2x}{\sqrt{9 - \cos^4 x}} dx$ OR $\int \frac{dx}{2x^2 - 5x + 3}$	2
25.	Solve: $\log\left(\frac{dy}{dx}\right) = ax + by$	
	Section:C	
Q26	Evaluate $\int \sqrt{\frac{x}{1-x^3}} dx$ OR $\int \frac{2x^2+3}{x^2(x^2+9)} dx$	
Q27.	For what value of k , the following function is continuous at $x = -\frac{\pi}{6}$ If $f(x) = \begin{cases} \frac{\sqrt{3}\sin x + \cos x}{x + \frac{\pi}{6}}, & x \neq -\frac{\pi}{6} \\ k, & \text{if } x = -\frac{\pi}{6} \end{cases}$	
Q28.	Find the particular solution of the equation $(x+1)\frac{dy}{dx} = 2e^{-y} - 1$ OR Solve: $\frac{dy}{dx} - 3\cot x = \sin 2x$	
Q29	Show that the function $f: R \rightarrow R$, given by $f(x) = \frac{x}{x^2+1}$ is neither one-one and nor onto.	
Q30.	If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & x \\ -2 & 2 & -1 \end{bmatrix}$ and $A \cdot A' = 9I$, A' is transpose of A. find x .	-2
Q31.	If $x^y + y^x = 1$, then find $\frac{dy}{dx}$	
	Section:D	
Q32.	If $(x-a)^2 + (y-b)^2 = c^2$, for some $c > 0$ prove that $\frac{[(1+y^2)^{\frac{3}{2}}]}{y^2}$ is constant. OR	

ARMY PUBLIC SCHOOL BARRACKPORE
HALF-YEARLY EXAMINATION(RETEST) 24-25
MATHEMATICS (041)

CLASS XII
SET C

TIME 3HRS

M.M-80

GENERAL INSTRUCTION:

1. ALL QUESTIONS ARE COMPULSORY

2. The question paper comprises of five sections- A, B, C, D & E. Each section is compulsory. However there are internal choices in some questions.

3. Section-A (1- 18) has 18 MCQs and (19 to 20) consists of 2 Assertion – Reason based questions of one mark each.

4. Section-B (21 to 25) has 05 VSA-type questions 2 marks each

5. Section-C (26 to 31) has 06 SA-type questions of 3 marks each

6. Section-D (32 to 35) has 04 LA-type questions of 5 marks each

7. Section-E (36 to 38) has source based/case-based units of assessment (4 marks each) with sub parts

Section A

Q1.	If A is A square matrix of order 2×2 such that $ A =2$, then the value of $ 4AA^{-1} $ is?	(a) 4	(b) 16	(c) 2	(d) 0
Q2.	If A be a skew symmetric matrix of order 3 and $ A = x$ then the value of $(2023)^x$ is	(a) $(2023)^0$	(b) $(2023)^3$	(c) 1	(d) 0
Q3.	Let $f: [2, \infty) \rightarrow R$, be the function defined by $f(x) = x^2 - 4x + 5$. The range of the function is	a) R	(b) $[4, \infty)$	(c) $[5, \infty)$	d) $[1, \infty)$
Q4.	The domain of the function $\cos^{-1}(2x - 1)$ is	a) $[0, 1]$	b) $[-1, 1]$	c) $(-1, 1)$	d) $[0, \pi]$
Q5.	The value of $\sin^{-1}(\sin(-\frac{17\pi}{8}))$ is	(a) $\frac{\pi}{4}$	(b) $\frac{\pi}{8}$	(c) $-\frac{\pi}{8}$	(d) $\frac{\pi}{2}$
Q6.	If $\frac{\pi}{2} < x < \pi$, find the value of x for which $\begin{bmatrix} 2\sin x & 3 \\ 1 & 2\sin x \end{bmatrix}$ is singular is	a) $\frac{\pi}{3}$	b) $-\frac{\pi}{3}$	c) $2\pi/3$	d) $\pi/2$
Q7.	The function $f(x) = \begin{cases} \frac{e^2 - e^{-x}}{2}, & x \neq 0 \\ k, & x = 0 \end{cases}$	a) 3/10	(b) -3	(c) 7	(d) none of these
Q8.	A function $f: R \rightarrow R$ defined by $f(x) = 2 + x^2$ is	Not Injective	b) injective	c) Surjective	d) Neither injective nor surjective
Q9.	$\int_0^{\pi} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$ is equal to	(a) -1	(b) $\frac{\pi}{2}$	(c) $\frac{\pi}{4}$	(d) $\frac{\pi}{6}$
Q10.	If $f(x) = \begin{cases} x^2, & x < 1 \\ 2 - x, & x \geq 1 \end{cases}$ then	a) not continuous at $x=1$	b) not differential at $x=1$	c) differentiable at $x=1$	d) neither continuous and not differentiable
Q11.	If $y = \sec(\tan^{-1}x)$ then, $\frac{dy}{dx}$ is equal to	a) $\frac{x}{\sqrt{1+x^2}}$	(b) $\frac{x}{1+x^2}$	c) $x\sqrt{1+x^2}$	(d) $-\frac{1}{\sqrt{1+x^2}}$
Q12.	If $y = Ae^x + Be^{-x}$ then $\frac{d^2y}{dx^2}$ is equal to	(a) $5y$	(b) $-y$	(c) y	(d) $25y$
Q13.	The interval in which the function $f(x) = x^2e^x$ is strictly increasing is	c) $(0, \infty)$	d) $(0, 2)$	a) $(-\infty, \infty)$	b) $(-\infty, 0)$
Q14.	The global minimum value of $f(x) = x^4 - x^3 - 2x + 6$ is	a) 6	b) 8	c) 4	d) does not exists
Q15.	If $f(x) = a(x - \cos x)$ is strictly decreasing in R the a belongs to	a) $(0, 1)$	b) $(0, \infty)$	c) $(-\infty, 0)$	d) $(-\infty, \infty)$
Q16.	If the function $f(x) = 2x^2 - kx + 5$ is increasing in $[1, 2]$, then k lies in the interval	a) $(-\infty, 4)$	b) $(4, \infty)$	c) $(-\infty, 8)$	d) $(8, \infty)$

Q34.	Find the area of the smallest region bounded by $x^2 + y^2 = 4$ and $x + y = 2$ OR Find the area of the region bounded by $\{(x, y): 0 \leq y \leq x^2 + 1, 0 \leq y \leq x + 1, 0 \leq x \leq 2\}$	
35.	If $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & -3 \\ -3 & 2 & -4 \end{bmatrix}$, find A^{-1} and hence solve the system of equations, $x + 2y - 3z = -4$, $2x + 3y + 2z = 14$, $3x - 3y - 4z = -15$	5
Section: E		
36.	A wire of length 34 cm is cut into two pieces. One of them into a square and another into a rectangle. The length of the rectangle is twice the breadth. If the length of the side of square be x a) Express the combined areas of square and rectangle in terms of x b) Find the lengths of the two pieces of the rod so that the combined area is minimum.	2+2
37.	The equation of the path traced by a roller coaster is given by $F(x) = a(x+9)(x+1)(x-3)$. If the path crosses y-axis at $(0, -1)$, then answer the following. i) Is the path is continuous. Justify ii) What is the value of a iii) Find the critical points of the function.	1+1 +2
38.	An organisation conducted a bike race under two different categories- boys and girls. There are total 28 participants in all. Among all of them, finally three from category-1 and two from category-2 two were selected for the final race. Let category-1, comprising boys be the set-A, and category-2 comprising girls be the set-B. $A = \{a, b, c\}$ and $B = \{p, q\}$. Based on the above information answer the following questions. a) How many relations are possible from A to B b) Among all possible how many functions can be formed from A to B. c) Let $R: A \rightarrow A$ be defined by $R: \{(x, y): x, y \text{ are students of same class}\}$. Check whether R is an	1+1 +2